

Branch: gh-pages genstats / inst / doc / 04_04_Gene_Sets.R

jtleek 28 days ago updated package

1 contributor

84 lines (68 sloc) 2.82 KB

RawBlameHistory

```
1  ## ----global_palette, results = 'asis'-----
2  rm(list=ls())
3  tropical= c('darkorange', 'dodgerblue', 'hotpink', 'limegreen', 'yellow')
4  palette(tropical)
5
6  ## ----global_options,warning=FALSE,message=FALSE-----
7  ## see ch. 10 Hooks of Xie's knitr book
8  library(knitr)
9  knitr_hooks$set(setPch = function(before, options, envir) {
10    if(before) par(pch = 19)
11  })
12  opts_chunk$set(setPch = TRUE)
13
14  ## ----global_plot,warning=FALSE, message=FALSE-----
15  knitr::opts_chunk$set(fig.width=5, fig.height=5, size="footnotesize",
16                        warning=FALSE, message=FALSE)
17  knitr::knit_hooks$set(small.mar = function(before, options, envir) {
18    if (before) graphics::par(mar = c(5,5,1.5,1))
19  })
20
21  ## ----load_hidden, echo=FALSE, results="hide", warning=FALSE-----
22  suppressPackageStartupMessages({
23    library(devtools)
24    library(Biobase)
25    library(goseq)
26    library(DESeq2)
27  })
28
29  ## ----load-----
30    library(devtools)
31    library(Biobase)
32    library(goseq)
33    library(DESeq2)
34
35  ## ----install_packages, eval=FALSE-----
36  # install.packages(c("devtools","MatrixEQTL"))
37  # source("http://www.bioconductor.org/biocLite.R")
38  # biocLite(c("Biobase","goseq","DESeq2"))
39
40  ## -----
41  head(supportedGenomes())
42  head(supportedGeneIDs())
43
44  ## -----
45  temp_data =read.table(system.file("extdata","Li_sum.txt",
46                                package="goseq"),sep="\t",
47                        header=TRUE,
48                        stringsAsFactors=FALSE)
49  expr= temp_data[,-1]
50  rownames(expr) = temp_data[,1]
51  expr = expr[rowMeans(expr) > 5,]
52  grp=factor(rep(c("Control","Treated"),times=c(4,3)))
53  pdata = data.frame(grp)
54
55  ## -----
56  de = DESeqDataSetFromMatrix(expr, pdata, ~grp)
57  de_fit = DESeq(de)
58  de_results = results(de_fit)
59
60  ## -----
61  genes = as.integer(de_results$padj < 0.05)
62  not_na = !is.na(genes)
63  names(genes) = rownames(expr)
```

https://github.com/jtleek/genstats/blob/gh-pages/inst/doc/04_04_Gene_Sets.R

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```
64 genes = genes[not_na]
65
66 ## -----
67 head(supportedGenomes(),n=12)[,1:4]
68
69 ## -----
70 pwf=nullp(genes,"hg19","ensGene")
71 head(pwf)
72
73 ## -----
74 GO.wall=goseq(pwf,"hg19","ensGene")
75 head(GO.wall)
76
77 ## -----
78 GO.MF=goseq(pwf,"hg19","ensGene",test.cats=c("GO:MF"))
79 head(GO.MF)
80
81 ## ----session_info-----
82 devtools::session_info()
83
```

